PRITHVIRAJ ARVIND PAWAR

Tampa, Florida, USA

+1-573-647-4395 pprithviraj18@gmail.com LinkedIn GitHub

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java, C++, C#, MATLAB, R.
- **AI/ML & Data Science:** Large Language Models (LLMs), RAG (Retrieval-Augmented Generation), LangChain, HuggingFace, Ollama, FAISS, Vector Databases, TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, Deep Learning, Computer Vision (CNNs), Natural Language Processing (NLP).
- **Web Frameworks:** Streamlit, Django, Flask, FastAPI, Node.js, React.js, Next.js, HTML5, CSS3.
- **Cloud & DevOps:** Cloud Computing (Azure), Containerization (Docker), CI/CD Pipelines, Git, Linux, Virtualization.
- **Databases:** PostgreSQL, MySQL, MongoDB, NoSQL, Apache Spark.

PROFESSIONAL EXPERIENCE

Solutions UI/UX — *Tampa, FL*

Software Developer *(Aug 2024 – Present)*

Software Developer Intern *(Jun 2023 – May 2024)*

- **Full-Stack Development:** Transitioned from prototyping mobile apps to architecting modular backend services using **Python** and **Node.js**. Built and optimized RESTful APIs integrated with React front ends.
- **Cloud Deployment:** Automated deployment workflows on Microsoft Azure using CI/CD pipelines and Docker, ensuring scalable and reliable application delivery for diverse clients.
- **Data & Databases:** Designed interactive Flask/Plotly dashboards for client insights and optimized MongoDB queries to improve application response times.

Netsol Network Consultancy Pvt. Ltd. — *Mumbai, India*

Junior Software Developer *(Jan 2021 – Mar 2022)*

Software Development Intern *(May 2020 – Dec 2020)*

- **Backend Engineering:** Developed robust backend services and REST APIs using **Python**, **Java**, and **SQL**, supporting high-traffic web applications.
- **Agile Collaboration:** Participated in daily stand-ups and code reviews, ensuring code quality and reducing post-production bugs through rigorous unit testing.
- **Cloud Support:** Assisted in the integration of databases and supported cloud deployment strategies, adapting quickly to full-stack workflows.

Missouri University of Science and Technology — *Rolla, MO*

Graduate Teaching Assistant

- Mentored 100+ students in Data Structures & AI algorithms; enhanced curriculum with inclusive teaching strategies.

PROJECTS

DocuChat AI: Local RAG System with Persistent Memory | *Python, LangChain, Ollama, Streamlit, FAISS*

- **Architected a Hybrid RAG Pipeline:** Designed a dual-mode chat system that switches seamlessly between General Knowledge (Llama 3) and Document Context (Retrieval-Augmented Generation) based on user interaction.
- **Engineered Persistence & State Management:** Solved complex session handling issues by implementing a file-based JSON storage system that retains multi-thread chat history and vector embeddings across application restarts.
- **Optimized for Cost & Privacy:** Eliminated OpenAI API costs by integrating Ollama (Llama 3) and HuggingFace Embeddings for local, offline inference on consumer hardware.
- **Implemented Vector Search:** Built a document ingestion pipeline using FAISS (Facebook AI Similarity Search) and Recursive Character Splitting to handle large PDF datasets with high retrieval accuracy.

Gendex: Gender Recognition using CNN [View Live Project]

- Trained a CNN-based gender recognition model with a diverse dataset of over 10,000 images, achieving 95% accuracy.
- Conducted real-time testing of the model with a dataset of 500 faces, achieving an impressive 92% accuracy in identifying gender.

BrandSuite: Analytics & Recommendation Engine | *Python, Azure, Docker, SQL*

- **Architected Scalable Backend:** Designed RESTful APIs on Azure to process brand performance data, replacing client-side analytics with robust server-side aggregation.
- **Built Recommendation Logic:** Engineered a ranking algorithm based on quality metrics and service performance, optimizing SQL queries for data consistency across distributed systems.
- **Containerized Deployment:** Implemented a CI/CD pipeline using Docker and Azure App Services, ensuring 99.9% system uptime and 30% faster API response times.

EdgeScope: Real-Time Object Detection Using Different Edge Detection Techniques

- **Published Research:** Benchmarked 10 Edge Detection algorithms, identifying a technique that improved Object Detection accuracy by 15% and processing speed by 20% [View Published Paper]

EDUCATION

Missouri University of Science and Technology — Aug 2022 - May 2024

Master of Science, Computer Science

- **Achievements:** Provost's Masters International Scholarship, CEC Deans Intl Masters Ac Scholarship

CERTIFICATIONS

- **Microsoft Certified: Azure Fundamentals**: Details available at: [View Credly Badge]